

KOHLER × MIT WORLD PEACE UNIVERSITY

AI Research Lab Program · Track 1

KOHLER AI Bathroom Designer & Planner

An AI-assisted constrained-recommendation system for complete bathroom configuration — from customer constraints to a feasible, budget-aware product bundle.

AI Understanding + Verified Retrieval + Optimization

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The Problem & The Objective

The Problem

Bathroom product selection is fragmented. A customer must understand dimensions and layout, identify what physically fits, compare specifications, match aesthetics, stay within budget, and check that chosen products are compatible with one another — a multi-constraint decision problem. Traditional product search treats products individually rather than solving the complete bathroom configuration as a system.

The Objective

Build an interactive AI design assistant that takes a customer's bathroom dimensions, layout/image, budget, design theme and product requirements, and automatically generates personalized, feasible, budget-aware KOHLER product bundle recommendations spanning faucets, smart toilets, thermostatic showers and vanities.

The variables that must be reasoned about together:

Dimensions &
Layout

Product
Compatibility

Budget

Design
Aesthetics

Functional
Requirements

Catalogue
Availability

In short — the proposed system converts a fragmented, multi-constraint search process into a single AI-assisted constrained recommendation problem.

Proposed Solution

An AI-powered bathroom-planning pipeline—not a product generator, a feasibility-optimizer.

1

User Constraint Collection

Dimensions, budget, theme & requirements from the customer

2

Bathroom Image / Layout Analysis

Spatial information extracted from an uploaded photo or layout

3

Product Catalogue Retrieval

Candidate KOHLER products retrieved from the vector store

4

AI-Based Reasoning

Requirements and space understood, constraints structured

5

Compatibility Checking

Candidates filtered for physical & specification fit

6

Constraint-Based Optimization

Budget, space & preference constraints applied together

7

Product Bundle Generation

A feasible, complete combination is assembled

8

Explanation of Recommendation

Reasoning returned along side the recommended bundle

Not aesthetics alone — the system generates a feasible and optimized product combination based on the customer's actual constraints — not simply attractive products.

Key Innovation — Separating Reasoning from Decisions

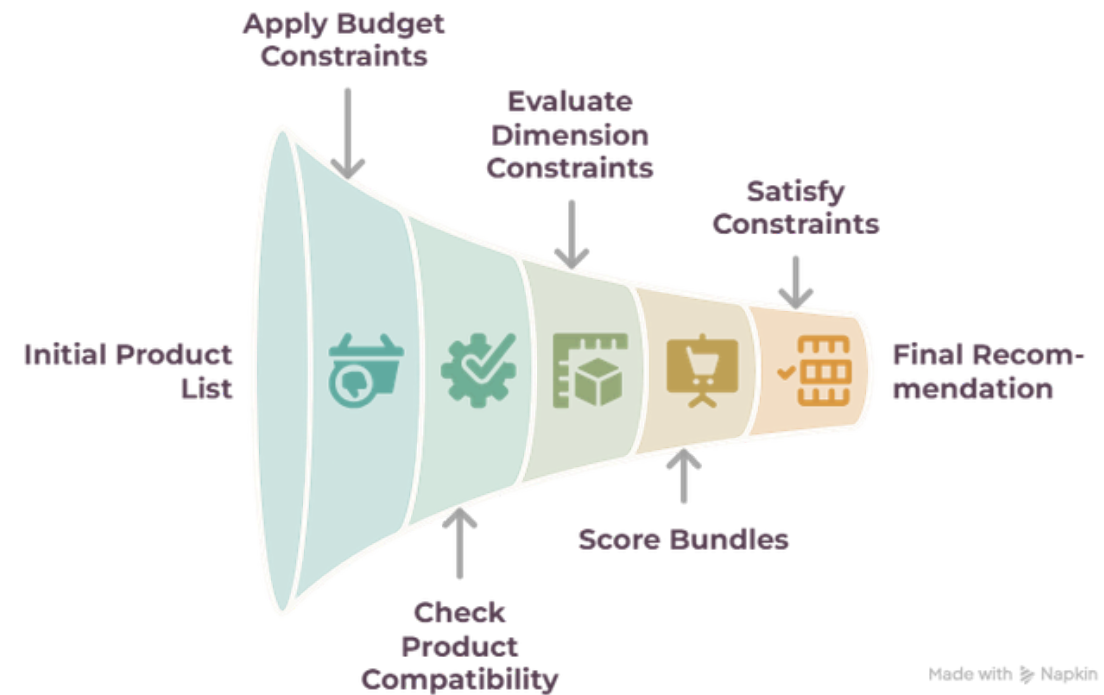
Enhancing Bathroom Design with AI



Made with Napkin

Deterministic / Optimization Handles

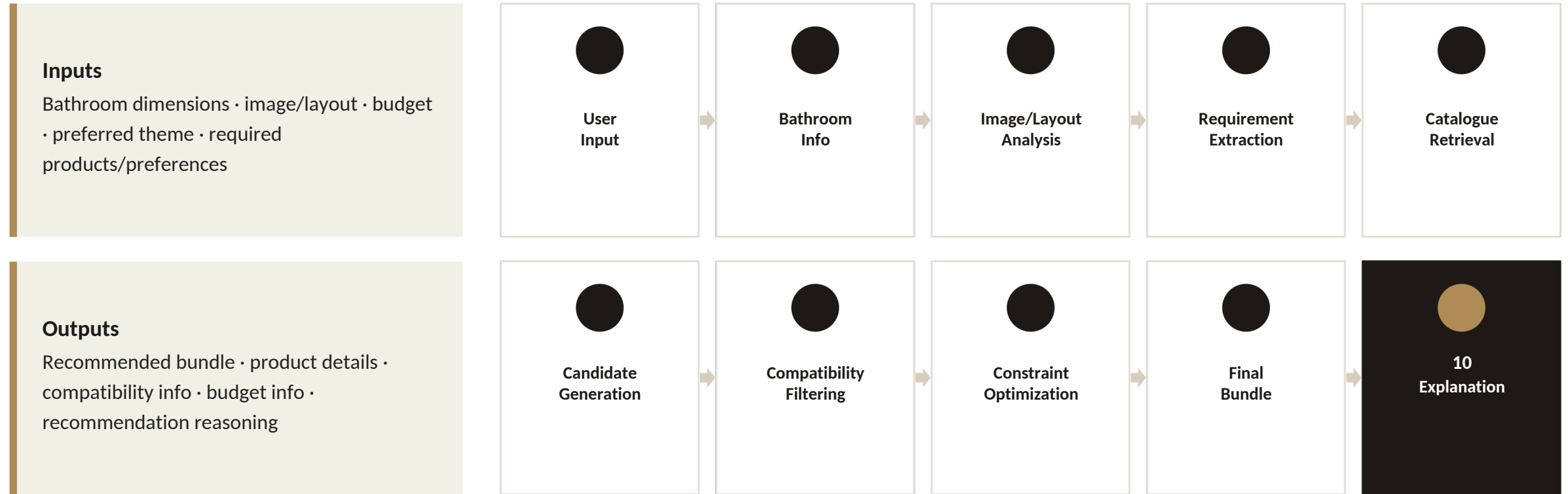
Product Recommendation Process



Made with Napkin

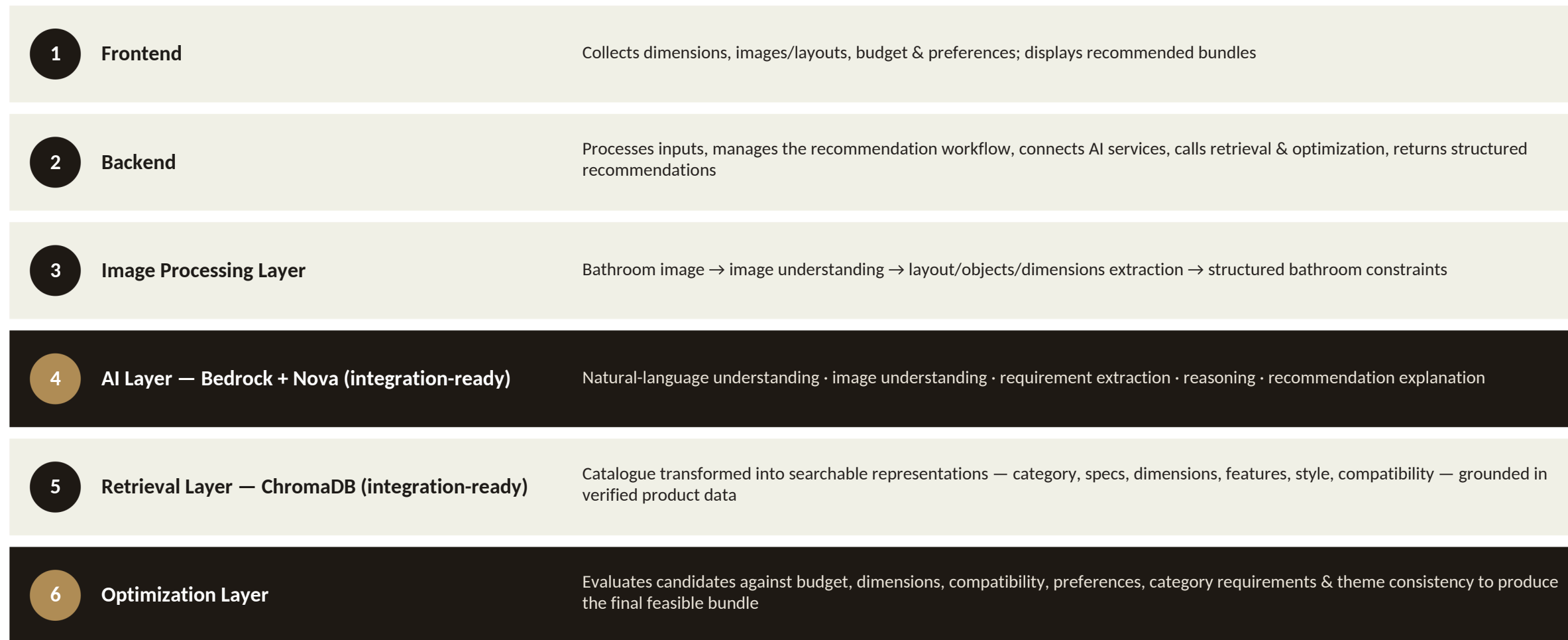
Why it matters — this reduces the risk of relying solely on generative AI for product-selection decisions.

User Workflow



End to end — ten steps carry the customer from raw inputs to an explained, feasible product bundle.

System Architecture



AI & Technology Stack

Frontend

React 19 • Vite 7 • TypeScript • Tailwind CSS • shadcn/ui — the bathroom-planning UI

Backend

Node.js • Express • tRPC • TypeScript • Zod • Drizzle ORM — type-safe API & orchestration

Data

MySQL / TiDB • S3-compatible storage • KOHLER catalogue — application & media persistence

AI & Intelligence

Amazon Bedrock / Nova (integration-ready) • Python / FastAPI • ChromaDB (integration-ready) • RAG

Core Engines

Catalogue retrieval • spatial intelligence • compatibility engine • colour intelligence • lookbook intelligence • constraint & budget optimization

DevOps & Testing

Docker • Vitest • Pytest • Git/GitHub

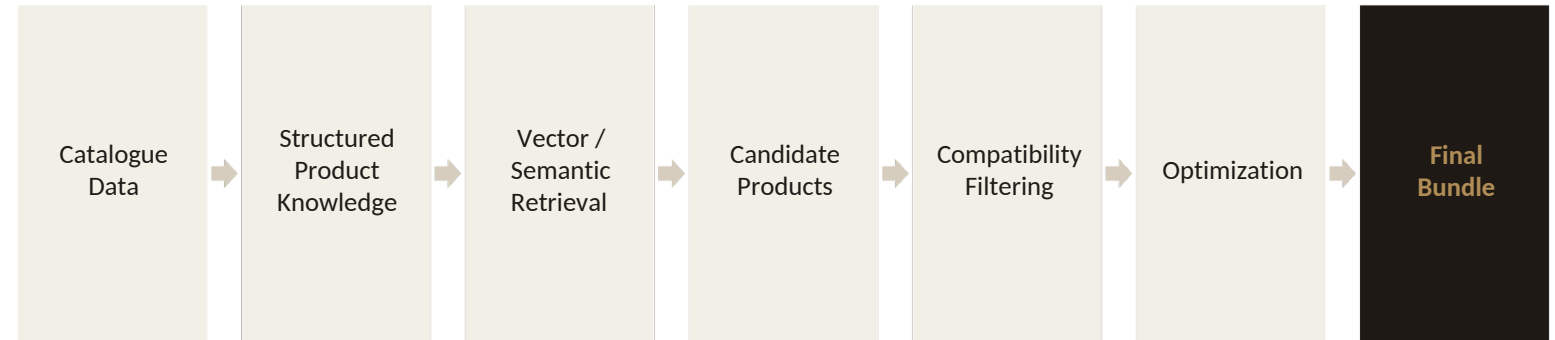
Grounding rule — the model is never allowed to invent product specifications — recommendations trace back to verified catalogue data.

Product Catalogue Intelligence

Attributes Retrieved

Product category
Dimensions
Technical specifications
Features
Installation requirements
Style / design attributes
Price / budget where available
Compatibility attributes

Retrieval→Decision Flow



The catalogue stays grounded in verified product information — retrieval identifies candidates before optimization ever runs.

Optimization Strategy

Objective—find a bathroom product bundle that maximizes overall suitability while satisfying user constraints.

Bundle Score = Compatibility + Preference Match + Design/Theme Match + Space Suitability + Feature Relevance – Budget Penalty – Constraint Violations

Hard Constraints — cannot be violated

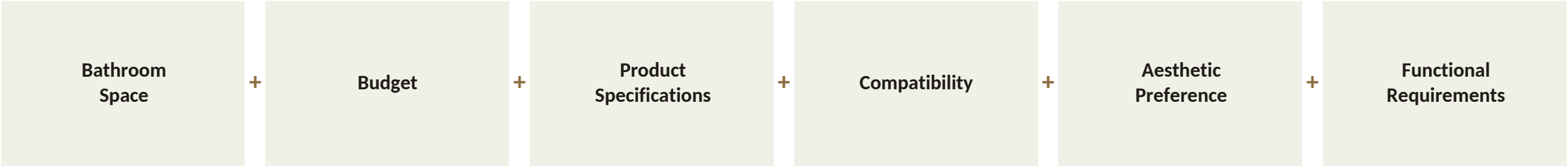
- Product must fit available space
- Required product categories must be satisfied
- Budget must not exceed the specified limit
- Products must satisfy compatibility requirements

Soft Constraints — can be traded off

- Preferred aesthetic theme
- Feature preferences
- Brand / product preferences
- Design coherence
- Additional convenience features

Why AI Is Required

A conventional product filter cannot reason across every one of these variables at once:



As product categories and constraints grow, the number of possible combinations increases rapidly.

AI understands — unstructured inputs — natural-language requirements, bathroom images, design preferences

Retrieval grounds — recommendations in verified catalogue data

Optimization decides — the structured selection problem

AI Understanding + Verified Retrieval + Optimization — three complementary capabilities, combined.

Indian Market Context & Differentiation

KOHLER operates in a competitive Indian bathroom & sanitaryware market.

Competitive landscape

Jaquar	Hindware
Cera	Parryware
Grohe	Roca

Context only — not a ranking or superiority claim. It establishes why an intelligent planning experience can differentiate.

Where this sits

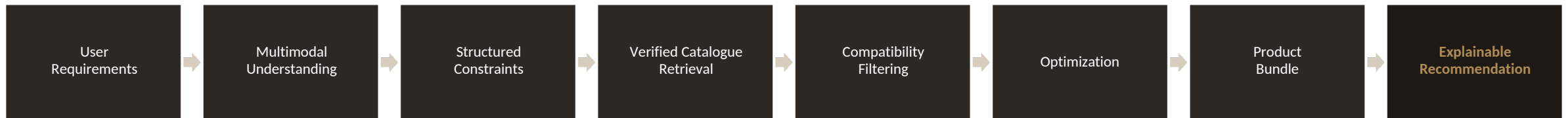
1. Traditional catalogue / product discovery
2. Product recommendation
3. AI-assisted bathroom planning
4. Constraint-aware complete-bundle recommendation

Technical Differentiation

Generic AI Chatbot



Proposed System

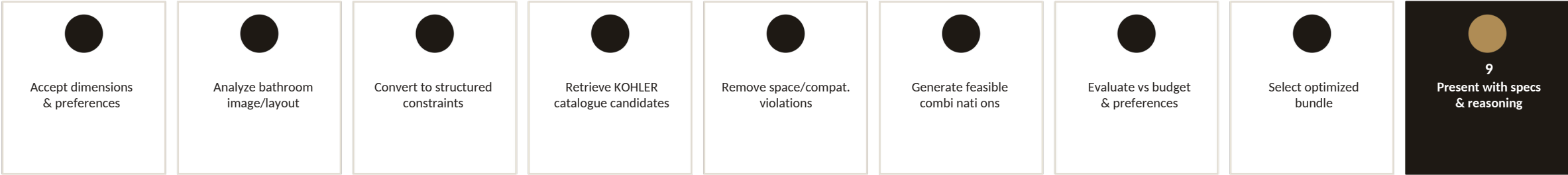


The proposed architecture combines generative AI with deterministic components rather than depending entirely on an LLM.

Example End-to-End Case

Bathroom size 8 ft × 6 ft	Budget ₹2,00,000	Theme Japanese Zen	Requirements Smart toilet, faucet, thermostatic shower, vanity
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System process



Demonstrates — a complete configuration problem being solved — not isolated product recommendations.

Feasibility & Business Value

Feasibility — modular by design

- Frontend for interaction
- Backend for orchestration
- Amazon Bedrock / Nova for AI
- Image processing for spatial understanding
- ChromaDB for retrieval
- Optimization engine for constraint satisfaction
- Structured catalogue data for product grounding

Business Value

- Reduces product-selection complexity
- Personalizes recommendations
- Helps customers understand suitable combinations
- Supports complete-bathroom configuration, not isolated discovery
- Connects requirements to relevant catalogue products
- Creates an AI-assisted digital planning experience

Cost-Efficient & Scalable AI — Amazon Nova 2 Lite vs. Claude Fable 5.1 (Anthropic API benchmark)

\$0.30 vs \$10.00 /M input

~97% lower input-token cost

\$2.50 vs \$50.00 /M output

~95% lower output-token cost

Text + Image + Video

native multimodal, matching bathroom-image inputs, on Bedrock

By using Amazon Nova 2 Lite through Amazon Bedrock, the solution combines multimodal AI with substantially lower token costs, improving the feasibility of scaling personalized bathroom design recommendations.

Scalability & Prototype Demonstration

Scalability — beyond the prototype

- Larger product catalogues • additional bathroom categories
- More detailed spatial reasoning • installation constraints
- Regional availability • price/availability updates
- More sophisticated optimization objectives
- Personalization from customer history
- Integration with digital commerce / showroom workflows

Prototype — what will be shown

- Enter dimensions, upload image, set budget & theme
- System retrieves relevant catalogue products
- Candidates filtered for hard constraints
- Optimization generates a feasible bundle
- Recommended products displayed with reasoning

Design principle — every component can be upgraded independently — the architecture doesn't need a redesign to grow.

THE FULL NARRATIVE

Problem → Opportunity → Solution → Innovation → Architecture
→ Methodology → Market Context → Example → Prototype → Scalability

Problem

Multiple interacting constraints

Opportunity

AI simplifies & personalizes planning

Solution

KOHLER AI Bathroom Designer & Planner

Innovation

Multimodal AI + retrieval + compatibility + optimization

Architecture

Frontend + Backend + Bedrock/Nova + Vision + ChromaDB + Optimization

Methodology

Retrieve candidates → apply constraints → optimize bundle

Market Context

Competitive Indian sanitaryware market

Example

One complete end-to-end planning scenario

Prototype

Working user workflow

Scalability

Evolves into a broader AI planning platform

From product search to intelligent bathroom configuration.